1 Discrete I: Electric Field due to an Electric Dipole (Parallel on the Dipole Axis)

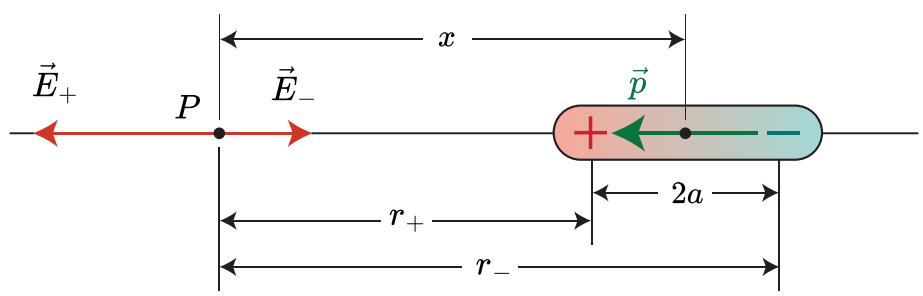

Figure 1: An electric dipole. The dipole's two charges result from the electric field vectors at point P on the dipole axis. Point P is at distances $r_+ = x - a$ and $r_- = x + a$ from the individual charges that make up the dipole. Here, $x \gg a$.

Measuring the field x distance away from the bisecting point of the dipole:

$$\begin{aligned}
\vec{E}_{\text{dipole}} &= \vec{E}_+ + \vec{E}_- \\
&= \frac{1}{4\pi\epsilon_0}\left(-\frac{q}{r_+^2} + \frac{q}{r_-^2}\right)\hat{i} \\
&= \frac{q}{4\pi\epsilon_0}\left\{\frac{1}{(x-a)^2} - \frac{1}{(x+a)^2}\right\}\hat{i} \\
&= \frac{q}{4\pi\epsilon_0}\left[\frac{(x-a)^2 - (x+a)^2}{\{(x+a)(x-a)\}^2}\right]\hat{i} \\
&= \frac{q}{4\pi\epsilon_0}\left[\frac{4xa}{(x^2-a^2)^2}\right]\hat{i} \\
&= \frac{q}{4\pi\epsilon_0}\left[\frac{2\cdot 2xa}{x^4}\right]\hat{i} \\
&= \frac{1}{4\pi\epsilon_0}\cdot 2\cdot\frac{q\cdot 2a}{x^3}\hat{i} \\
&= \frac{1}{4\pi\epsilon_0}\frac{2p}{x^3}\hat{i} \\
&= \frac{1}{2\pi\epsilon_0}\frac{p}{x^3}\hat{i}.
\end{aligned} \tag{1}$$

The field points from the direction of the dipole moment. i.e., toward the positive charge.

To account for the arbitrary direction of the observation point P along the dipole axis. We replace x with r, giving us:

$$\vec{E}_{\text{dipole}} = \frac{1}{2\pi\epsilon_0}\frac{\vec{p}}{r^3}\hat{r}_{(-)\to(+)} = \frac{1}{2\pi\epsilon_0}\frac{p}{r^3}\hat{p} \tag{2}$$

The electric field of a dipole thus points in the direction of the electric dipole moment, regardless of whatever orientation we try. It further establishes the *why* behind the convention of $\vec{p}$'s direction.

2 Discrete II: Electric Field due to an Electric Dipole (Perpendicular to the Dipole Axis)

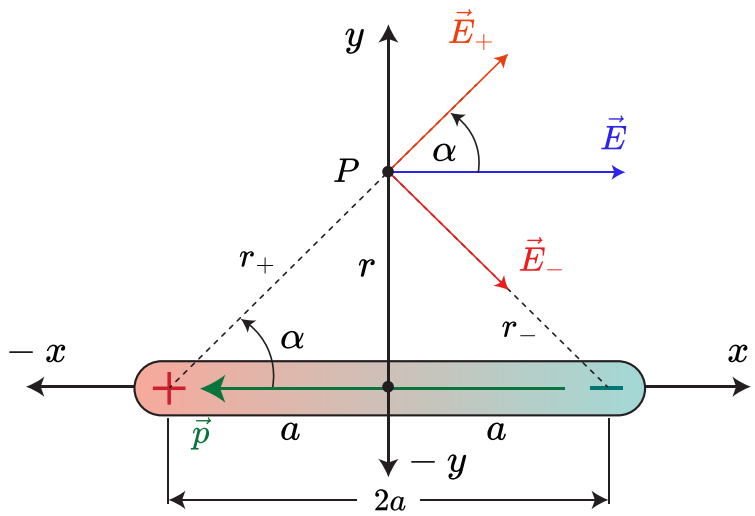

Figure 2: An electric dipole produces an electric field at point P off the dipole axis. Point P is at a distance r perpendicularly off the dipole axis. Here, $r \gg a$.

Measuring the field r distance, perpendicularly away from the bisecting point of the dipole:

$$\begin{aligned}
\vec{E}_{\text{dipole}} &= \vec{E}_+ + \vec{E}_- \\
&= \{E_+(x) + E_-(x)\}\,\hat{i} + \{E_+(y) + E_-(y)\}\,\hat{j} \\
&= \{E_+(x) + E_-(x)\}\cos\alpha\hat{i} + 0\hat{j} \\
&= \frac{2}{4\pi\epsilon_0}\cdot\frac{q}{y^2+a^2}\cdot\frac{a}{\sqrt{y^2+a^2}}\hat{i} \\
&= \frac{q\cdot 2a}{4\pi\epsilon_0}\frac{1}{(y^2+a^2)^{\frac{3}{2}}}\hat{i} \\
&= \frac{1}{4\pi\epsilon_0}\frac{p}{(y^2+a^2)^{\frac{3}{2}}}\{-\hat{p}\}\,. \qquad (3)
\end{aligned}$$

The field points from the direction of the dipole moment. i.e., toward the positive charge.

To account for the arbitrary direction of the observation point P along the dipole axis. We replace y with r, giving us:

$$\vec{E}_{\text{dipole}} = \frac{1}{4\pi\epsilon_0}\frac{p}{(r^2+a^2)^{\frac{3}{2}}}\{-\hat{p}\} \qquad (4)$$

In the limit $r \gg a$, we get the following:

$$\vec{E}_{\text{dipole}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}\{-\hat{p}\}\,. \qquad (5)$$

We can see that Eqs. (5) and (2) are identical except for a factor of two, which shows up only when the field is measured along the dipole axis.

In both cases, it is evident that the electric field due to an electric dipole falls off as $\frac{1}{r^3}$. This is called the *Inverse Cube Law*, unlike the well-known *Inverse Square Law* we encountered for a single-point charge.